

13. | Implement your own ping program.

AIM: To implement our own ping program in Python that resolves a host name to its IP address and measures the response time of the host.

Procedure

1. Open a Python file **ping_program.py** and import the *socket* and *time* modules.
2. Read the host name or IP address from the user.
3. Resolve it to an IP address with *socket.gethostbyname()*; if this fails, print "Invalid host/IP address."
4. Repeat 4 times: note the start time, open a TCP connection to port 80 of the host using *socket.create_connection()* with a 2-second timeout, and note the end time.
5. Print the reply with the response time in milliseconds $((\text{end} - \text{start}) \times 1000)$ and close the socket. If the connection fails, print "Request timed out."
6. Wait 1 second between requests, then run `python ping_program.py` and enter a host name.

ping_program.py

```
import socket

import time

host = input("Enter host/IP address: ")

try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")

    for i in range(4):
        start = time.time()

        try:
            s = socket.create_connection((ip, 80), timeout=2)

            end = time.time()

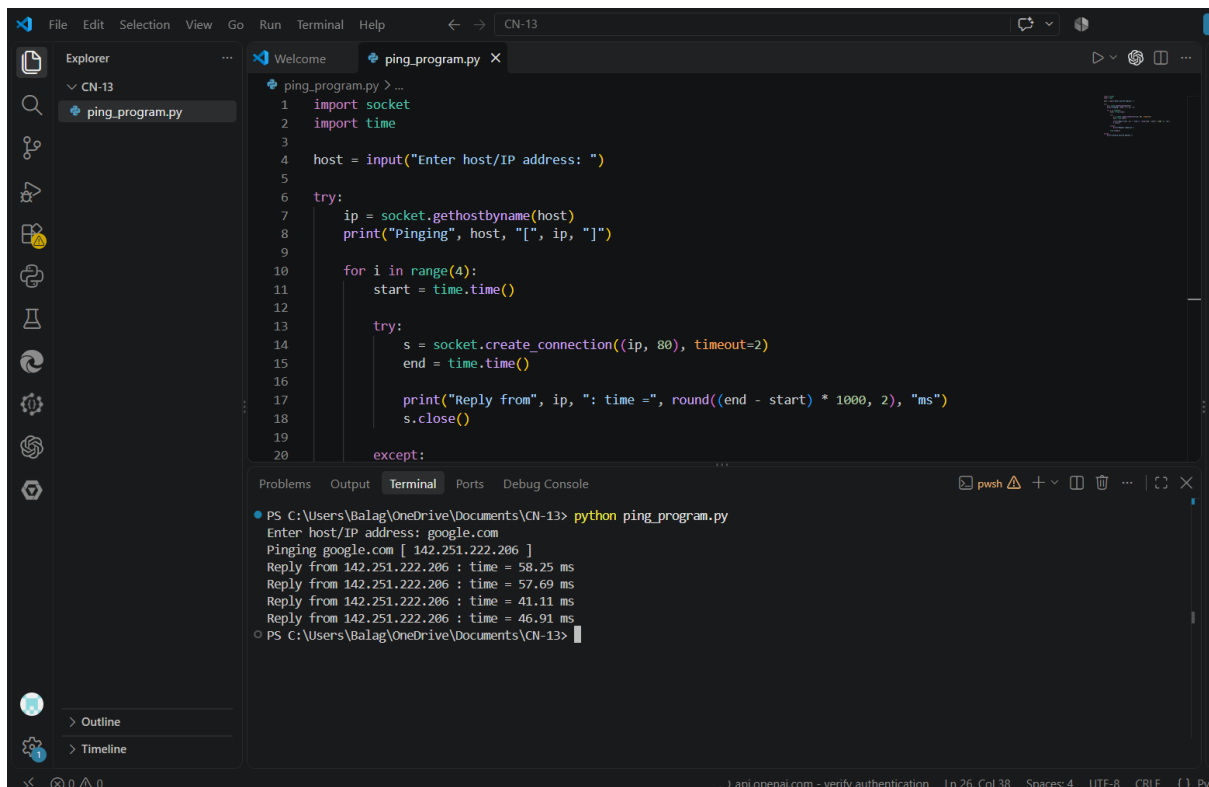
            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")

            s.close()
```

```
except:  
  
print("Request timed out.")
```

```
time.sleep(1)
```

```
except:  
  
print("Invalid host/IP address.")
```



The screenshot shows the Visual Studio Code editor with a file named `ping_program.py` open. The code is a Python script that takes a host/IP address as input and performs a ping test. It uses the `socket` module to create a connection and the `time` module to measure the response time. The script is executed in the terminal, and the output shows the ping results for `google.com`.

```
1 import socket  
2 import time  
3  
4 host = input("Enter host/IP address: ")  
5  
6 try:  
7     ip = socket.gethostbyname(host)  
8     print("Pinging", host, "[", ip, "]")  
9  
10    for i in range(4):  
11        start = time.time()  
12  
13        try:  
14            s = socket.create_connection((ip, 80), timeout=2)  
15            end = time.time()  
16  
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")  
18            s.close()  
19  
20 except:
```

Terminal Output:

```
PS C:\Users\Balag\OneDrive\Documents\CN-13> python ping_program.py  
Enter host/IP address: google.com  
Pinging google.com [ 142.251.222.206 ]  
Reply from 142.251.222.206 : time = 58.25 ms  
Reply from 142.251.222.206 : time = 57.69 ms  
Reply from 142.251.222.206 : time = 41.11 ms  
Reply from 142.251.222.206 : time = 46.91 ms  
PS C:\Users\Balag\OneDrive\Documents\CN-13>
```

```
● PS C:\Users\Balag\OneDrive\Documents\CN-13> python ping_program.py  
Enter host/IP address: google.com  
Pinging google.com [ 142.251.222.206 ]  
Reply from 142.251.222.206 : time = 58.25 ms  
Reply from 142.251.222.206 : time = 57.69 ms  
Reply from 142.251.222.206 : time = 41.11 ms  
Reply from 142.251.222.206 : time = 46.91 ms  
○ PS C:\Users\Balag\OneDrive\Documents\CN-13>
```

Result

The ping program was implemented in Python. The host name `google.com` was resolved to the IP address `142.251.222.206`, and four replies were received with response times of 58.25 ms, 57.69 ms, 41.11 ms and 46.91 ms.